

SynCocreate: Fostering Interpersonal Connectedness via Brainwave-Driven Co-creation in Virtual Reality

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ABSTRACT

Collaborative art and co-creation enhance social wellbeing and connectivity. However, the combination of art creation through mutual brainwave interaction with the prosocial potential of EEG biosignals reveal an untapped opportunity. SynCocreate presents the design and prototype of a VR-based interpersonal electroencephalography (EEG) neurofeedback co-creation platform. This generative VR platform enables paired individuals to interact via brainwaves in a 3D virtual canvas, painted and animated collaboratively through their real-time brainwave data. The platform employs synchronized visual cues, aligned with the real-time brainwaves of paired users, to investigate the potential of collaborative neurofeedback in enhancing co-creativity and emotional connection. It also explores the use of Virtual Reality (VR) in fostering creativity and togetherness through immersive, collective visualizations of brainwaves.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Collaborative interaction**;

KEYWORDS

Generative VR, Interpersonal social connectedness, Co-creation, EEG

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1 INTRODUCTION

The exploration of collaborative artwork and co-creation in human interactions is extensive, ranging from the classic exquisite corpse game to traditional forms like visual art, musical composition, and choreography [1]. These practices are known to foster social wellbeing such as enhancing cooperation [2] and reinforcing bonds [3]. However, collaborative art creation based on mutual brainwave interaction remains relatively underexplored. Meanwhile,

measurable biosignals, such as EEG, relay internal activities and emotional states. Sharing and synchronizing biosignals such as brainwaves bear prosocial potential for interaction [4], yet expressing and sharing these signals, especially in remote social contexts, remains challenging.

In SynCocreate, by harnessing the collaborative potential of neurofeedback, we offer a playful and engaging experience for co-creating artwork within a VR environment through synchronizing brainwaves between pairs of users. The experience unfolds within a dome that symbolizes the brain. Two spectral, smoke-like sprites dance in the space, each guided by the real-time brainwave signals of a user. As the users' brainwaves synchronize, a splash of paint materializes on the dome's surface like a firework. Each user's environment incorporates visual cues — such as light, color, and pulsations synchronized with their paired partner's brainwaves — to encourage closer brainwave alignment. With each synchronization, more paint is added, gradually creating a vibrant, unique painting on the dome's interior, resembling a sky filled with fireworks, celebrating their connection.

In this paper, we describe the main related work, the design and implementation details, and the interactive experience, and discuss the potential value of the SynCocreate platform and future work.

2 RELATED WORK

We were inspired by previous work in co-creation using biometric data, including the EMS Painter that uses electrical muscle stimulation to influence a painter's movements [5], and WiiArts that employs body gestures for joint art creation [6].

Potentially closer to our work is research that explores the connection between EEG signals, emotions, and co-performance. [7] demonstrates that sharing emotions between individuals through EEG can effectively foster social connectedness, intuitive understanding, and empathy. [8] suggests that the experience of reaching for neural synchronization can amplify feelings of social connection. [9] finds improved music quality when participants' brain signals are synchronized. These studies suggest that synchronizing interpersonal brainwaves enhances both emotional experiences and collaborative endeavors.

We also draw upon studies that explore the impact of external visual stimuli on modifying brainwave ratios and improving an individual's focus [10], and that highlight the potential for brainwaves to synchronize with a range of external frequencies, encompassing visual, auditory, and tactile modalities [11].

SynCocreate differentiates from prior work by focusing on collaborative art creation driven by mutual brainwave interaction and synchronization. It employs gamified interactive elements and leverages VR visual stimuli to enhance brainwave synchronization

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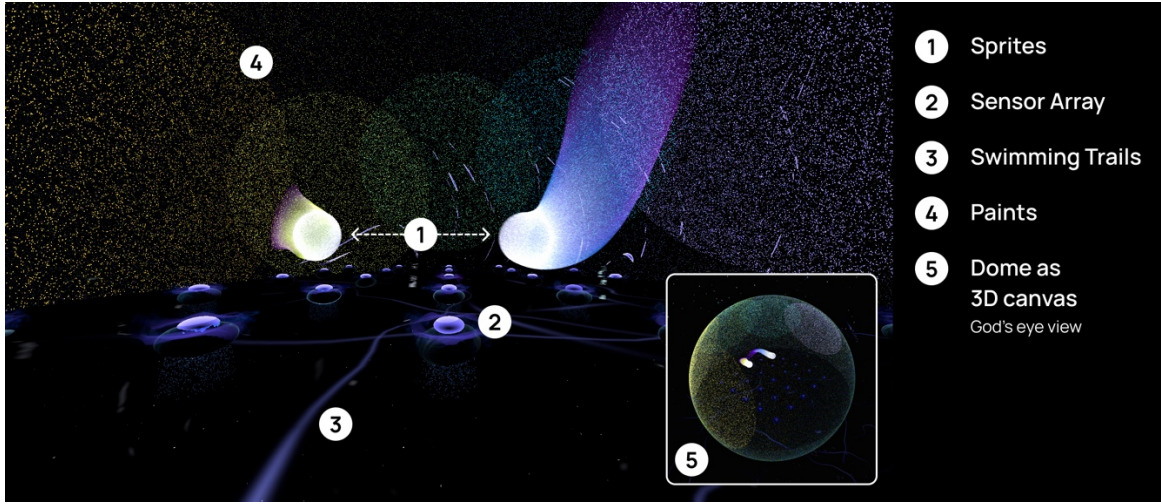


Figure 1: Annotated scene elements.

and foster emotional connections in the realms of co-creation and co-performance.

3 DESIGN AND IMPLEMENTATION

Imagine a patient with early signs of dementia, coupled with social anxiety and mistrust towards a new caregiver. They step into this immersive starry brain dome together and observe as their neural patterns collaboratively paint the dome by synchronizing their brainwaves. This experience offers them a unique opportunity to express their mind states in a controlled, creative environment, fostering emotional comfort through non-verbal interaction. As they witness the visual harmony created by their joint brain activity, they find a new way to connect and understand each other beyond words.

Using Muse headbands - a lightweight, commercially available brainwave sensing device - we capture each user's brainwave data in real-time, and process and import the real-time band types, ratios and frequencies into Unity via an OSC server and the built-in WebSocket library. The Muse headband uses a default sampling frequency of 10Hz. There are five band types based on common classification of brainwave frequency bands defined by their frequency ranges: Gamma (γ) (above 30 Hz), Beta (β) (14 - 29 Hz), Alpha (α) (8-13 Hz), Theta (θ) (4-7 Hz) and Delta (δ) (1-3 Hz) [12]. In the Unity environment, we map these real-time brainwave data from both users to CG properties within the scene to craft a dynamic visual representation of brain activities.

3.1 Scene Design

The scene is set in a starry dome arena designed to resemble the hemispherical shape of the brain, transporting users into an imaginative cognitive realm. This space is complemented by a rippling, reflective water surface, on which an array of sensors floats, metaphorically serving as conduits for streaming real-time data from an external source. Luminescent trails swim beneath the water, symbolizing transient thoughts. Above the water under the dome, two

ethereal sprites, each representing a user's brain activity, dance dynamically (Figure 1).

Departing from conventional two-dimensional art canvases, the hemispherical dome serves as a three-dimensional canvas for collaborative artwork driven by the users' brainwave data. This approach serves to reveal the subtleties of the users' brainwave activities and provides a tangible representation of their cognitive processes. The collaborative interactive experience is crafted with predefined rules connecting specific aspects of brainwave data to various elements within the 3D environments.

3.2 Data Mapping

Drawing inspiration from established psychological color narratives, where bright, warm colors evoke activeness and dim, cool colors induce calmness, we developed a color mapping scheme. This scheme transitions from yellow to purple counterclockwise along the color wheel, representing brainwave band types from γ to δ , aligning with brain activity and mental states in order of decreasing activeness.

In each user's environment, the pulsation frequency of the sensors is driven by the real-time frequency of the other user's dominant brainwave. The primary color of the scene, which is applied to both the sensors and the swimming trails, reflects the other user's dominant brainwave type. These elements act as visual cues, aiming to enhance the synchronization of the users' neural rhythms. The tail color and movement vector of each sprite corresponds to each user's own dominant brainwave band type, providing a visual representation of their mental state in comparison to their partner's.

Previous studies have identified distinct patterns linking brainwave bands to specific brain regions: positive emotions typically activate the β and γ bands in temporal areas, while negative emotions often result in heightened δ band activity in occipital regions [13]. Similarly, [14] noted variations in EEG power across emotional states induced by VR videos, emphasizing the α -band's significant role in the prefrontal region for positive emotions compared to

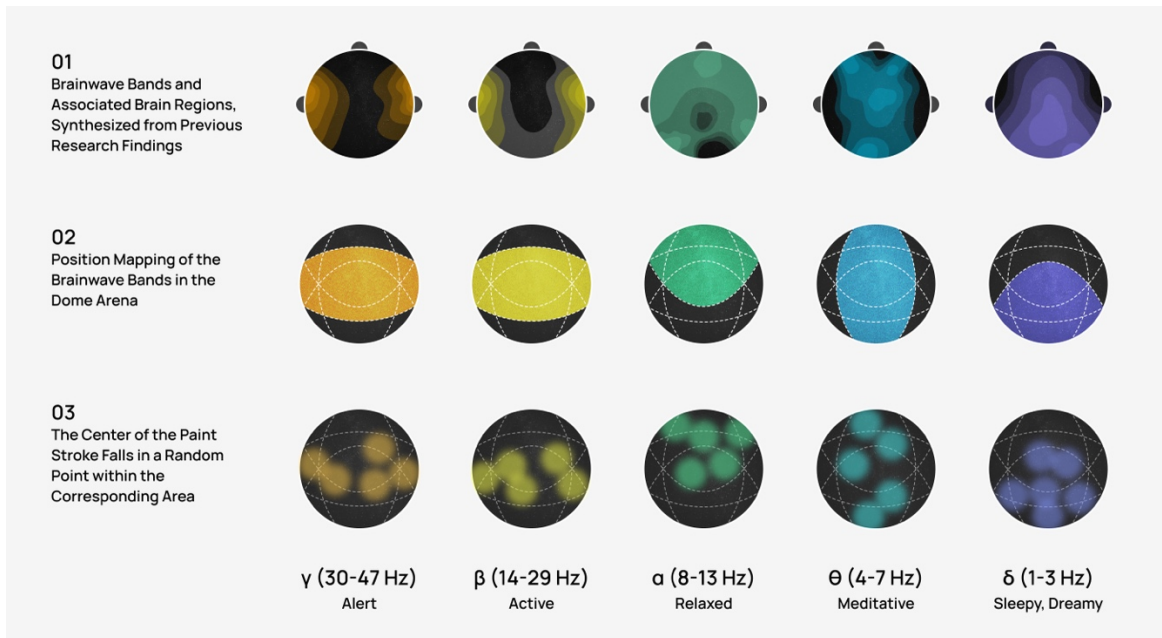


Figure 2: Position mapping diagrams of the brainwave bands in the dome arena, plan view.

negative ones. In our scene, the mapping of sprite and paint stroke positions is an artistic interpretation of these findings, harmonized with symmetry rules, and adjusted to ensure overlaps among the areas. Drawing on the relationship between brainwave band types and their primary brain activation regions, we have associated the β and γ bands with the brain's middle latitudinal area, the θ band with the middle longitudinal area, the α -band with the frontal area, and the δ band with the rear area (Figure 2).

3.3 Movement Logic and Painting Mechanism

The two sprites dynamically shift positions based on the dominant brainwave type and its corresponding brain dome area. For example, when a user's dominant band type is α , a random point within the predefined prefrontal area is selected, and the sprite moves gradually towards it. When both users synchronize in the δ band, their sprites simultaneously head towards a mutual point in the rear area. Each synchronization event between the paired users results in a grainy ink stroke, colored according to the corresponding band type, appearing in the relevant area of the dome (Figure 3). The more frequent their synchronization, the more paint accumulates on the hemispherical canvas, creating overlapping artistic effects as they synchronize across different band types. To aid the synchronization process, each pair of users is encouraged to follow the other's trail within the scene.

4 INTERACTIVE EXPERIENCE

SynCcreate offers an immersive and interactive collaborative experience in a 3D art environment, inviting pairs of users – be they friends, couples, colleagues, or strangers – to connect creatively via brainwaves. As users stream their brainwaves into the system, they

learn to collaboratively activate paints within the dome, crafting imaginative artworks in a brain-inspired space (Figure 4).

The experience is enriched by two primary layers of interactivity. First, the interaction allows one user's brainwaves to dynamically modify the other's environment, enhancing mutual awareness and empathy. Users can choose to converse, exert internal effort, or simply let the visual cues guide them towards synchronization. Second, as the session progresses, the blending paints coalesce into a 'neural painting', resembling a dome of fireworks in the sky, celebrating the users' neural resonance and connection. This evolving virtual art piece creates moments of eager anticipation, with each harmonious brainwave alignment marked by vibrant splashes of paint.

5 DISCUSSION

Our preliminary user testing with the existing prototype shows SynCcreate's ability to boost creativity and foster a sense of togetherness without verbal communication. We found that the platform's gamified experience prompts users to purposefully observe the mental state of their counterpart and synchronize collaboratively to produce unique artwork infused with mutual brainwave signals. Many users expressed interests in internalizing brainwave classifications and color mapping for more active pattern tracking. While the rules are generally user-friendly, some might require more time to become familiar with them. To respond to this interest and support the learning process, we've introduced a choice of default or alternative color palettes, like choosing a paint palette kit before starting. Future plans include a popup legend card and customizable mappings to enhance personalization. The experience has also generated discussions on its broad applicability in various contexts, from enhancing cognitive functions and emotional

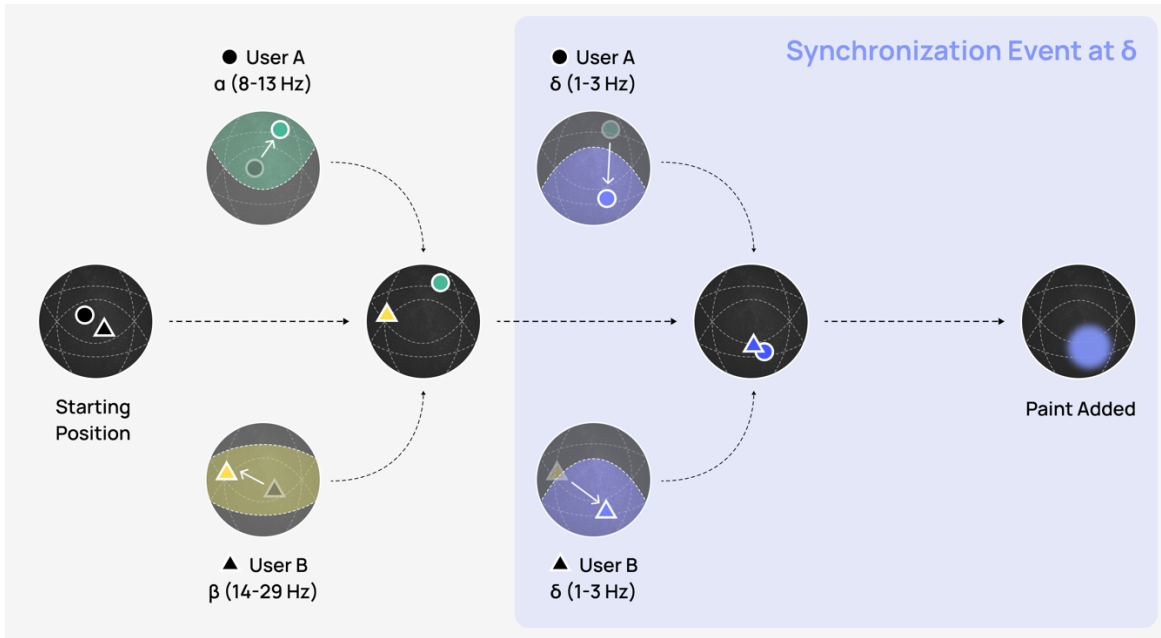


Figure 3: Movement logic example diagram, plan view.

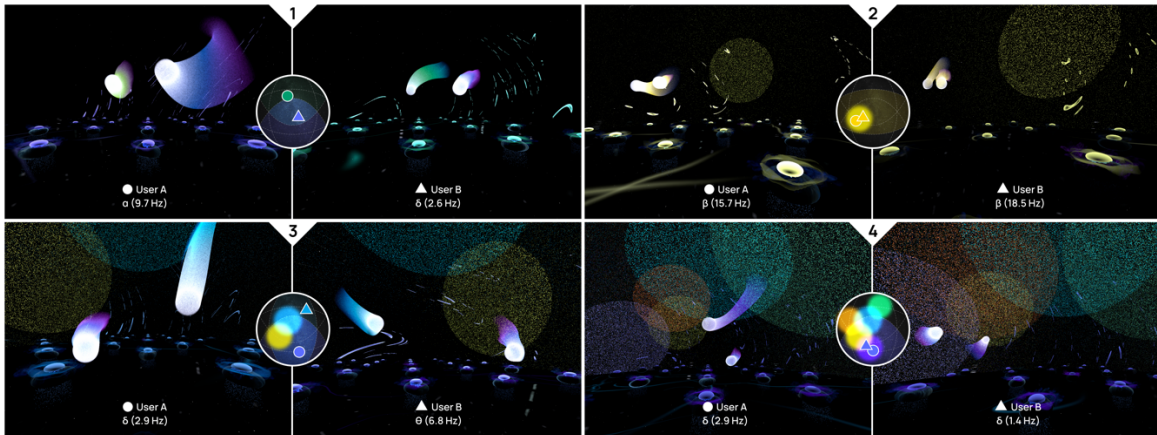


Figure 4: Examples of artwork generated based on various synchronization events between users.

comfort in elderly care, to serving as a therapeutic tool in mental health, fostering collaboration in educational and corporate settings, bridging cultural gaps, and enabling artistic collaborations.

The project introduces a novel co-creation approach by combining virtual platforms with biometric data synchronization, offering fresh perspectives on interaction modalities and interpersonal co-creation for the ACM community. This fusion of design and technology creates a visually appealing, tangible, and engaging experience. It encourages contemplation on how combining bio-data, emerging technologies, and experience design can enhance non-verbal interpersonal interactions, leading to greater accessibility and inclusivity. Additionally, the project aims to demonstrate

how HCI can contribute to emotional and social wellbeing, enriching holistic wellness through unique experiences in collaboration, coordination, and creativity.

Looking ahead, we hope to gain deeper insights into the ways in which creativity is amplified by EEG data, and further develop the platform's potential to enable groups of individuals to synchronize biosignals and collaboratively create artworks in real-time.

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